

LANGLEY RESEARCH CENTER

FACILITY LOCATION Hampton, Virginia 23665
FACILITY NUMBER 1247-B
FACILITY NAME 22-Inch Helium Tunnel
FUNCTIONAL NAME Wind Tunnel, Hypersonic Helium, 22-In.
TECHNOLOGICAL AREAS Aerodynamic forces and moments, pressure distribution, heat transfer, flow visualization

INITIAL COST	\$ 997 K	YR. BUILT	1960	STATUS CODE	Active
ACCUM. COST	\$ 1,312 K	NASA B.O.D.	1960	OWNER CODE	NASA
LIFE EXPECT.	Indef.			OPER. CODE	NASA

CONTRACTOR NAME
(if contr. oper.)

POTENTIAL

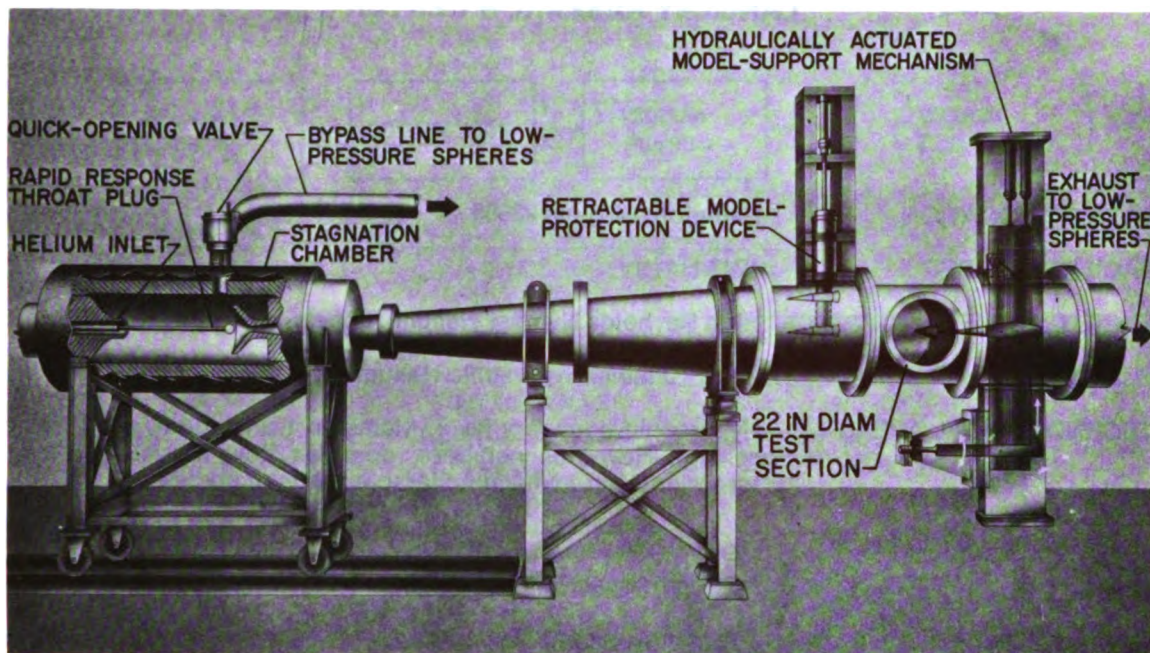
PLANS

OTHER INFO SOURCES Longitudinal Characteristics of Several Configurations at Hypersonic Mach Numbers in Conical and Contoured Nozzles, NASA TN D-2489, September, 1964

COGNIZANT ORG. Space Systems Division
COMPONENT

LOCAL CONTACT FOR Chief, Research Facilities Engineering Division, Code 56.000; (804)
FURTHER INFO 827-3171

January 1974



DESCRIPTION

The test medium, purified helium, is normally supplied at ambient temperature (400°F maximum). The tunnel is a blowdown-type requiring about 1.5 hr between runs to evacuate the gas-collection system. Run times are normally 30 sec with a limit near 60 sec. Models are mounted on a vertical strut with an angle-of-attack range of $\pm 18^\circ$; bent stings extend the range. Sideslip angles are set at constant values for sweeps through the angle-of-attack range. Force and moment data are obtained at 14 discrete angles over a 36° -angle range during a run. An electron beam illuminates the flow for 3-dimensional flow-visualization studies. The test section is 22.5 in. in diameter, and the test core is 8 to 10 in. in diameter. Model size is limited to approximately 6 in. for angles up to 60° and 12 in. up to 30° . Three contoured nozzles with design Mach numbers of 18, 22, and 26 are available; however, only the Mach 22 nozzle has been calibrated.

CHARACTERISTICS (Mach 22 Nozzle)

Stagnation Pressure, psia: 200 to 3000

Mach Number: 17.6 to 22.2

Reynolds Number, per ft: 1.1×10^6 to 11.3×10^6

Maximum Running Time, sec: 60 to 40

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FACILITY LOCATION Hampton, Virginia 23665
FACILITY NUMBER 1247-H
FACILITY NAME High Reynolds Number Helium Tunnels
FUNCTIONAL NAME Wind Tunnels, Hypersonic Helium
TECHNOLOGICAL AREAS Hypersonic boundary layers (laminar, transitional, and turbulent) at high Reynolds numbers; complex flow fields and flow interactions; aerodynamic characteristics of bodies

INITIAL COST	\$ 3,148 K	YR. BUILT	1967	STATUS CODE	Active
ACCUM. COST	\$ 3,148 K	NASA B.O.D.	1967	OWNER CODE	NASA
LIFE EXPECT.	Indef.			OPER. CODE	NASA

CONTRACTOR NAME
(if contr. oper.)

POTENTIAL

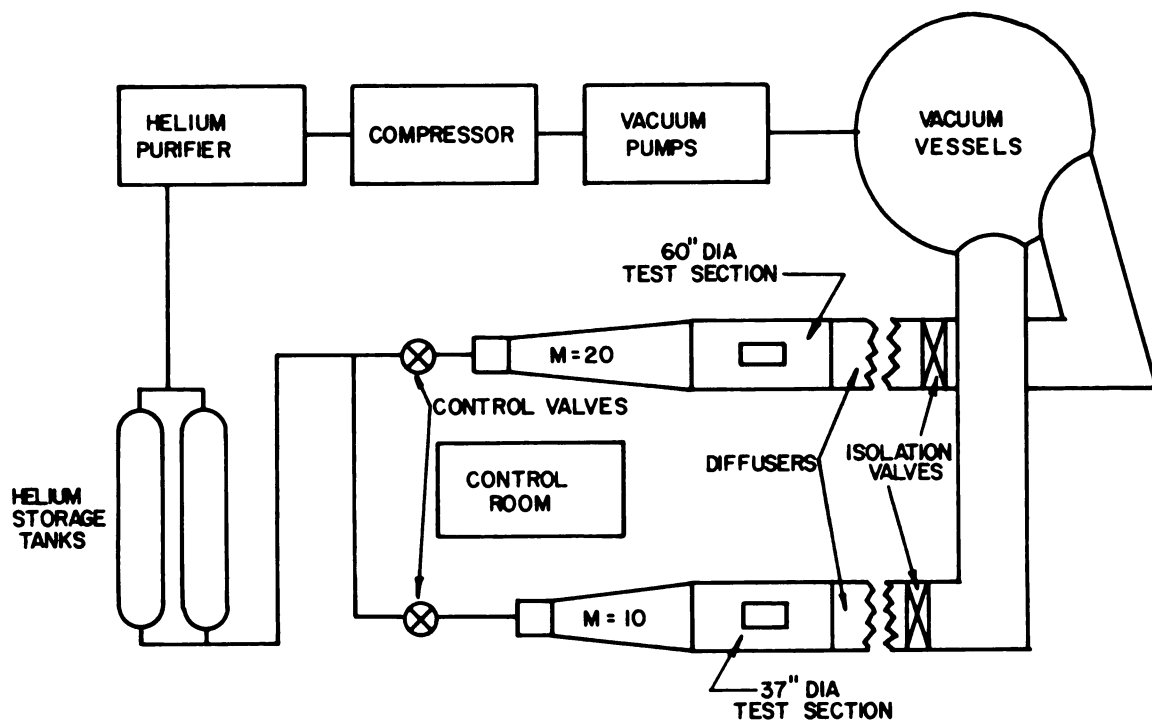
PLANS

OTHER INFO SOURCES Operating Characteristics of the Langley Mach 10 High Reynolds Number Helium Tunnel, NASA TM X-2955; Calibration of the Langley Mach 20 High Reynolds Number Helium Tunnel Including Diffuser Measurements, NASA TM X-2353, October, 1971

COGNIZANT ORG. COMPONENT Hypersonic Vehicles Division

LOCAL CONTACT FOR FURTHER INFO Chief, Research Facilities Engineering Division, Code 56.000; (804) 827-3171

January 1974



DESCRIPTION

The tunnels consist of Mach 10 and Mach 20 legs connected to a common high-pressure helium supply and vacuum system. The test medium is helium at ambient temperature. Models can be sting and strut mounted. If they are sting mounted, the maximum model length is about 5 ft for the Mach 20 leg and 3 ft for the Mach 10 leg. The angle of attack can be changed during a run. Test-section diameters are 60 in. for the Mach 20 leg and 37 in. for the Mach 10 leg. Core diameters are 20 in. for each leg. Each tunnel has spark schlieren capability.

CHARACTERISTICS

Mach Number:	10	20
Velocity, ft/sec:	5880	5950
Maximum Stagnation Pressure, psia:	4000	4000
Stagnation Temperature, °R:	540	540
Maximum Dynamic Pressure, psia:	54.6	7.2
Unit Reynolds Number, per ft:	60.2×10^6	20.4×10^6
Maximum Running Time, sec:		5